Practical 3

In [1]:
```
import tarfile
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Define a filter function that allows only regular files to be extracted
def tar_safe_filter(tarinfo, path=None):
    if tarinfo.isreg(): # Check if it's a regular file
        return tarinfo # Include this file in extraction
    else:
        return None # Skip non-regular files
    # Open and extract the .tgz file using the filter argument
        with tarfile.open("cal_housing.tgz", "r:gz") as tar:
            tar.extractall(path=".", filter=tar_safe_filter)

# Load dataset from the extracted file
california_data = pd.read_csv("cal_housing.data", header=None, delimiter=",")

# Assign column names
california_data.columns = [
"MedInc", "HouseAge", "AveRooms", "AveBedrms", "Population",
"AveOccup", "Latitude", "Longitude", "MedHouseVal"
]

# Separate features and target variable
X_reg = california_data.drop('MedHouseVal', axis=1)
y_reg = california_data['MedHouseVal']

# Split the data for training/testing
X_reg_train, X_reg_test, y_reg_train, y_reg_test = train_test_split(X_reg, y_reg, test_size=0.3, random_state=42)
```

```python
# Initialize and train the model
linear_reg = LinearRegression()
linear_reg.fit(X_reg_train, y_reg_train)

# Predict and evaluate
y_reg_pred = linear_reg.predict(X_reg_test)
mse = mean_squared_error(y_reg_test, y_reg_pred)
r2 = r2_score(y_reg_test, y_reg_pred)
print(f"Multiple Linear Regression - MSE: {mse}, R2 Score: {r2}")
```

```
Multiple Linear Regression - MSE: 4757467642.769163, R2 Score: 0.6375373510914579
```